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Participatory temporal urbanism: Reimagining participatory design, place evolution, and sustainable cities

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Abstract

Public participation has long occupied a central position within planning and design theory. From Arnstein's seminal work on citizen participation to collaborative planning, co-production, and contemporary participatory design approaches, scholars and practitioners have increasingly recognized the importance of involving citizens and users in shaping urban environments. Despite these advances, participation continues to be understood predominantly as an episodic and project-based activity associated with specific interventions, workshops, or consultation processes. Relatively limited attention has been devoted to the temporal dimensions of participation and to how users, places, meanings, and needs evolve over time. This study argues that time remains one of the least-explored dimensions of participatory planning and design. Building on literature from planning theory, participatory design, commons scholarship, care studies, and emerging work on temporality, the study proposes participatory temporal urbanism as a conceptual framework for understanding participation as an ongoing and evolving relationship rather than as a discrete event. The framework introduces five interrelated dimensions: co-creation, deliberation, co-management, co-adaptation, and co-memory. Drawing on concepts such as stewardship, commoning, maintenance, and care, the study further argues that sustainable urban development requires forms of participation that extend beyond implementation and continue throughout the life cycle of places. Particular attention is given to deliberative approaches, including the jury principle associated with New Urbanism, and to the role of participatory design in accommodating changing social, cultural, and intergenerational needs. By introducing time as a central analytical category, the study contributes to ongoing debates on participatory planning and participatory design and offers a new framework for understanding the collective processes through which sustainable cities can be collectively shaped, maintained, and continuously reinterpreted over time.

Keywords: Participatory planning; Participatory design; Temporal urbanism; Place evolution; Sustainable cities; Stewardship; Commons; Urban design

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Citation: Haas T. Participatory temporal urbanism: Reimagining participatory design, place evolution, and sustainable cities. *Design+*. 2026;3(3):026270030. doi: 10.36922/DP026270030

Received: June 29, 2026

Revised: July 11, 2026

Accepted: July 22, 2026

Published online: July 31, 2026

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1. Introduction

Public participation has become one of the most influential and enduring ideas in planning and design theory. Over the past several decades, participatory approaches

have evolved from early debates concerning citizen power and advocacy planning to collaborative planning, co-production, participatory design, and more recent forms of deliberative and digital engagement.[1-4] Across these traditions, the involvement of citizens, communities, and users has increasingly been regarded as essential for democratic legitimacy, social inclusion, and sustainable urban development.

At the same time, contemporary cities face increasingly complex, uncertain, and dynamic challenges. Climate change, demographic transitions, changing lifestyles, technological transformations, and evolving patterns of everyday life continuously reshape how people use, experience, and attach meaning to places. Urban environments are not static objects but living and evolving settings whose meanings and functions change over time. As Jacobs[5] famously observed, cities derive their vitality from the countless interactions and everyday practices that unfold within them. More recently, Sennett[6] has argued that cities should be understood as open and adaptive systems capable of accommodating change and diversity over time.

Despite the richness and diversity of participation scholarship, much of the literature continues to conceptualize participation as episodic and project-based. Public hearings, workshops, charrettes, consultations, and co-design processes are often associated with specific planning interventions and with limited timeframes. Participation is frequently treated as something that occurs before implementation, during a project, or at specific decision-making moments. Comparatively less attention has been devoted to what happens after projects have been completed, how places continue to evolve, and how users participate in the long-term adaptation, stewardship, and reinterpretation of urban environments.

Recent scholarship on care, commons, maintenance, and urban stewardship has begun to challenge purely project-oriented understandings of planning and design. Mattern,[7] for example, emphasizes the importance of maintenance and repair as fundamental yet often overlooked dimensions of urban life. Similarly, de la Bellacasa[8] has highlighted the significance of care as an ethical and relational practice extending beyond conventional approaches to management and governance. These perspectives suggest that cities are not simply built and completed but continuously maintained, adapted, and reinterpreted.

At the same time, emerging scholarship on temporality has highlighted the importance of rhythms, acceleration, waiting, and unequal temporal experiences in shaping contemporary societies.[9,10] Yet these temporal perspectives have rarely been systematically connected to debates on participatory planning and participatory design. Time, despite its importance for urban life, remains one of the least-explored dimensions of participation theory.

Despite the remarkable development of participatory planning and design over the past five decades, participation research remains theoretically fragmented. Different traditions have examined participation through distinct conceptual lenses, including citizen participation, collaborative planning, participatory design, co-production, stewardship, commons governance, care, maintenance, adaptive management, and post-occupancy learning. While each has generated valuable insights, these perspectives have largely evolved independently and have rarely been integrated through a common conceptual framework.

This study argues that time represents the missing integrative dimension of participation theory. Rather than proposing another model of participation, participatory temporal urbanism (PTU) is introduced as an integrative conceptual framework that reconnects these previously separate strands through a shared temporal perspective. PTU conceptualizes participation as an ongoing relationship between people, places, institutions, and urban change that unfolds throughout the life cycle of places rather than ending with project implementation.

The study offers three principal contributions. First, it synthesizes previously fragmented debates on participatory planning, participatory design, collaborative planning, stewardship, commons governance, care, maintenance, and temporality into a coherent conceptual framework. Second, it establishes time as a central analytical category for understanding participation, place evolution, and sustainable urban development. Third, it introduces PTU as an integrative conceptual framework that extends existing approaches by conceptualizing participation as a continuous process embedded within the long-term life of places rather than as a series of isolated planning events.

The remainder of the article is structured as follows. Section 2 presents the methodological approach. Section 3 reviews the evolution of participatory planning and design and discusses the emergence of care, commons, and temporal perspectives. Section 4 examines the historical development of participatory approaches. Section 5 introduces the notion of participatory temporal design (PTD), while Section 6 presents the framework of PTU. Section 7 illustrates the framework through selected examples, followed by a discussion (Section 8) and concluding reflections (Section 9).

2. Methodological approach

This study adopted a conceptual research design based on an integrative review and theoretical synthesis of literature across planning, urban design, participatory design, governance, commons scholarship, care studies, and temporal theory. Rather than undertaking a systematic literature review, the objective was to bring together influential and complementary bodies of scholarship that have generally developed in parallel and examine them through a common temporal perspective.

The literature was selected based on its theoretical relevance to participatory planning and design, collaborative governance, stewardship, maintenance, adaptive urbanism, and temporality. Priority was given to foundational and widely cited contributions that have significantly shaped debates within their respective fields, while more recent scholarship was incorporated to capture emerging perspectives on care, commoning, maintenance, adaptive governance, and temporal urbanism.

The PTU framework emerged through an iterative process of conceptual analysis, comparison, and synthesis. Existing theories were examined to identify common themes, conceptual overlaps, and areas where temporal dimensions remained underdeveloped. Based on this analysis, the five interrelated dimensions of PTU, co-creation, deliberation, co-management, co-adaptation, and co-memory, were developed as an integrative framework for understanding participation as a continuous relationship unfolding across the life cycle of places.

3. State of the art: Participatory planning and design

Participation has occupied a central position in planning and design theory for more than half a century. Throughout this period, understandings of participation have evolved considerably, reflecting changing views concerning democracy, expertise, citizenship, and the relationship between users and the built environment. While different traditions have emphasized distinct dimensions of participation, they collectively reflect an increasing recognition of citizens and users as active contributors to urban development. The following sections review five broad strands of scholarship that have shaped contemporary understandings of participation and provide the foundation for the framework proposed in this study.

3.1. Participation and power

Modern debates on participation emerged in response to technocratic and expert-driven planning traditions, including the rational-comprehensive model and its incrementalist critics.[11] Early advocates of participation argued that planning decisions should not be left exclusively to professionals and public authorities but should instead involve citizens directly. Advocacy planning, associated with Davidoff,[12] emphasized representation and pluralism in planning processes. Similarly, Friedmann[13] argued for planning approaches that empower communities and support social learning.

Underlying these debates is a more fundamental question of what power itself entails. Lukes[14] distinguished three dimensions of power: (i) its visible exercise in open decision-making; (ii) its less visible operation in controlling agendas and suppressing issues before they reach debate; and (iii) its most subtle form, which shapes preferences and perceptions so that conflict never surfaces at all.

Participatory planning theory has tended to focus on the first of these dimensions, treating participation primarily as a corrective to exclusion from formal decision-making. Yet Flyvbjerg's[15] ethnographic study of planning practice in Aalborg demonstrated that power frequently operates beneath the surface of formal participation procedures, shaping which rationalities and arguments are even considered legitimate. Planning processes that appear procedurally open can therefore still reinforce existing power asymmetries, a tension that has remained central to participation theory ever since.

Arnstein's[1] influential "ladder of citizen participation" remains one of the most widely cited contributions to participation theory. Her framework highlighted the distinction between genuine participation and tokenistic forms of consultation, emphasizing participation as a question of power and citizen control. The ladder's enduring influence, however, has not gone unchallenged. Tritter and McCallum[16] argued that its single-axis focus on power obscures other important dimensions of participation, including the diversity of knowledge that participants bring and the value of the participatory process itself, and proposed a "mosaic" model in its place. Yiftachel[17] extended this critical line further, arguing that planning has historically functioned not only as a vehicle for empowerment but also as an instrument of social control, particularly toward marginalized groups.

These critiques established that participation cannot be assumed to be inherently progressive; its effects depend on how power is distributed, exercised, and concealed within specific institutional and political contexts. Together, these contributions established the normative foundations of participatory planning while setting the terms for an enduring debate about the relationship between participation and power that continues to shape contemporary practice.

3.2. Communicative and collaborative planning

Beginning in the 1980s and 1990s, participation theory increasingly shifted from questions of power and conflict toward communication, dialogue, and consensus-building. Influenced by deliberative democracy and theories of communicative action, planning scholars began to challenge technocratic understandings of expertise and instead to emphasize interaction, negotiation, and collective learning as central elements of planning practice.

Forester[18] argued that planners should be understood not merely as technical experts but also as mediators, facilitators, and communicators capable of addressing conflict and fostering mutual understanding. Similarly, Healey[2] developed collaborative planning as an approach that emphasizes inclusive dialogue, shared knowledge, and institutional capacity-building. In this perspective, planning becomes a process through which different actors negotiate meanings, values, and future visions rather than merely producing technical solutions.

In parallel, Innes and Booher[3] highlighted the importance of collaborative rationality and argued that participation can generate new forms of knowledge through interaction among diverse stakeholders. Their work emphasized consensus-building, social learning, and network governance as essential dimensions of contemporary planning processes. Likewise, Cars[19] demonstrated how negotiation and conflict management could provide constructive mechanisms for addressing competing interests within planning and land-use decisions.

Collectively, these approaches represented a major shift away from hierarchical, expert-driven models toward more communicative, relational understandings of planning. Participation increasingly came to be understood not only as a mechanism for redistributing power but also as a process through which trust, legitimacy, social learning, and shared understanding could emerge. Rather than treating citizens as passive recipients of expert knowledge, collaborative planning traditions recognized communities as active participants in the co-production and shaping of urban futures.

Although communicative and collaborative planning have been criticized for underestimating power asymmetries and structural inequalities[15] and for relying excessively on consensus-oriented assumptions that may overlook persistent conflicts and competing rationalities,[20] they nevertheless represent one of the most influential paradigmatic shifts in modern planning theory and continue to shape contemporary approaches to participation, governance, and democratic engagement. Their emphasis on dialogue, social learning, negotiation, and institutional capacity-building has profoundly influenced subsequent developments in participatory planning, collaborative governance, co-production, and participatory design.

3.3. Participatory design and co-production

Parallel developments occurred within design disciplines. Participatory design traditions originating in Scandinavia emphasized the active involvement of users in shaping technologies, workplaces, and built environments.[21] In architecture and urban design, Sanoff[4] promoted community participation as a means of improving design quality and strengthening democratic engagement. More recent work on co-design and co-production has further challenged traditional distinctions between experts and users. Scholars such as Sanders and Stappers[22], Manzini[23], Björgvinsson *et al.*[24] and argue that citizens should be regarded not merely as stakeholders but as creative participants capable of contributing knowledge, experience, and innovation. These perspectives have broadened understandings of participation beyond consultation toward collaboration and shared authorship.

Design disciplines introduced additional modes of participation. Community design workshops and participatory design approaches sought to integrate users more directly into creative processes.[4,21] Similarly, New Urbanism popularized the charrette as an intensive and collaborative design process bringing together professionals, stakeholders, and citizens over relatively short timeframes. The charrette tradition has been particularly associated with collaborative urban design and accelerated community planning.[25,26] Within the New Urbanist tradition, Duany *et al.*[27] emphasized participatory approaches to reconnect design expertise with citizen engagement.

As summarized in Table 1, participation has expanded from questions of power and representation toward collaboration, co-production, care, and, more recently, temporal perspectives.

3.4. Care, commons, and stewardship

Recent scholarship has increasingly emphasized maintenance, repair, care, and stewardship as essential yet often overlooked dimensions of urban life. Ostrom's[28] work on commons governance demonstrated how communities can collectively manage shared resources over long periods, challenging the assumption that shared resources inevitably degrade due to overuse or neglect. Subsequent work by Bollier[29] expanded these ideas through the concept of commoning, understood not as a fixed institutional

Table 1. Evolution of participatory approaches in planning and design

Period	Dominant ideas	Typical methods	Representative thinkers
1950s–1960s	Rational and public-interest planning	Public hearings, public meetings	Charles E. Lindblom and Paul Davidoff
1960s–1970s	Participation and redistribution of power	Community meetings, citizen advocacy	Sherry R. Arnstein
1980s–1990s	Communicative and collaborative planning	Workshops, consensus building	Patsy Healey, John Forester, and Judith E. Innes
1990s–2000s	Participatory design and New Urbanism	Charrettes, co-design exercises	Henry Sanoff, Pelle Ehn, and Andrés Duany
2000s–2010s	Co-production and commons	Living labs, community partnerships	Elinor Ostrom and Ezio Manzini
2010s–2020s	Care, stewardship, and digital participation	Digital platforms, citizens' assemblies	Shannon Mattern and María Puig de la Bellacasa
Participatory temporal urbanism	Participation across time	Co-creation, deliberation, co-management, co-adaptation, and co-memory	Tigran Haas, Henri Lefebvre, Barbara Adam, and Hartmut Rosa

arrangement but as an ongoing social practice through which communities continuously negotiate access, responsibility, and use.

In parallel with this commons literature, scholars working on maintenance and repair have challenged understandings of urban development that privilege construction and innovation over the unglamorous, continuous work required to keep places functioning. Mattern[7] has argued that cities depend not only on innovation and infrastructure but also on ongoing maintenance and repair practices, work that is frequently rendered invisible precisely because successful maintenance prevents visible disruption. de la Bellacasa[8] further highlights care as an ethical and relational practice that extends beyond conventional management approaches, reframing care not as a residual or secondary activity but as a constitutive element of how human and more-than-human worlds are sustained collectively.

Parallel insights have emerged from environmental governance. Research on adaptive governance[30] and adaptive co-management[31] demonstrates how institutions can learn, adjust, and share responsibility for complex social-ecological systems over long periods through iterative cycles of feedback and adjustment. Although developed primarily in relation to natural resources, these concepts are directly relevant to urban public space, where conditions, users, and demands likewise change in ways that cannot be fully anticipated by any initial plan.

A closely related strand of research has translated this concern a direct urban-design vocabulary. Dempsey and Burton[32] distinguish place-keeping from place-making, defining place-keeping as the long-term management of public spaces and showing that capital continues to be invested in creating high-quality places without adequate resources for maintaining them afterward. Their work demonstrates that place-keeping depends on partnerships, sustained funding, and governance arrangements that extend well beyond project delivery, an insight that, from the management side, anticipates the temporal argument this study develops from the perspective of participation.

More recently, this body of thinking has been extended directly into urban design and planning. Gabauer *et al.*[33] bring together cross-disciplinary accounts of how care and uncare are unevenly produced and distributed across urban space, showing that caring relations are shaped by inequality, infrastructure, and everyday political struggle rather than occurring automatically wherever communities are present. Davis[34] develops this further into an explicit ethics of care for urban design, arguing that the built environment itself can be shaped to support relations of interdependence, capability, and long-term wellbeing, rather than treating care as something added to places after design is complete. Together, these contributions suggest that care is not merely a value to be invoked in planning rhetoric but a practice that can, and arguably should, be designed for.

Collectively, these perspectives challenge project-oriented understandings of urban development and emphasize continuity, responsibility, and stewardship as ongoing achievements rather than fixed outcomes of a single planning or design intervention. They also reframe

the temporal assumptions of participation discussed throughout this study: if the long-term life of places depends on sustained maintenance, repair, and care, then participation, too, cannot be confined to discrete moments of consultation and decision-making but must extend into the continuing work of keeping places alive.

3.5. Time as the missing dimension

Alongside these developments, scholars from sociology, geography, mobility studies, and urban studies have increasingly examined the temporal dimensions of contemporary life. Lefebvre's[35] rhythmanalysis, Adam's[36] work on timescapes, Rosa's[9] theory of social acceleration, Sharma's[10] work on temporal politics, and Sheller's[37] contributions to mobility justice all highlight the importance of time in shaping social experiences and inequalities. Despite these advances, temporal perspectives have rarely been systematically integrated into participatory planning and participatory design. Existing scholarship has generated important insights into power, collaboration, co-production, care, and stewardship. Yet participation itself is still largely understood as an episodic activity associated with discrete projects and interventions. Moreover, relational perspectives on urban life emphasize that cities are continuously produced through everyday interactions, encounters, and evolving social relations rather than fixed physical forms.[38] Such perspectives further reinforce the need to understand participation as an ongoing process rather than a singular event. The temporal dimensions of participation, including how users, places, meanings, and needs evolve over time, remain comparatively underexplored.

Collectively, these traditions have significantly enriched participatory planning and design. However, while existing scholarship has expanded understandings of power, collaboration, co-production, and care, relatively limited attention has been devoted to how participation unfolds across time. This omission forms the basis for the framework proposed in the following sections.

4. Participation methods and practices

While participatory planning and design have generated extensive theoretical debates, they have also produced a wide variety of practical methods and tools. Over time, planners, designers, and communities have experimented with different forms of engagement intended to improve democratic legitimacy, strengthen local knowledge, and foster collective ownership of urban development processes. These methods reflect changing understandings of participation and illustrate the increasing diversification of participatory practices.

Early participatory approaches were largely based on public hearings, public meetings, and consultation processes organized by planning authorities. Such methods emerged from traditions of representative democracy and public administration and were primarily intended to inform citizens and gather feedback on proposed plans. Although these approaches increased transparency, critics frequently argued that they provided limited opportunities for meaningful influence and often reproduced existing power asymmetries.[1]

The rise of advocacy planning during the 1960s and 1970s encouraged more active forms of community engagement. Citizen groups, neighborhood organizations, and local activists increasingly became involved in planning processes, particularly regarding issues of social justice and representation.[12] In the following decades, workshops, visioning exercises, and collaborative forums became common tools in communicative and collaborative planning traditions.[2,18] More recently, co-design practices, living laboratories, digital platforms, and citizens' assemblies have expanded the repertoire of participatory methods and enabled broader forms of engagement.[23]

In parallel, tactical and temporary urbanism have introduced explicitly temporally conscious practices into this repertoire,[39,40] deploying short-term, low-cost interventions to test ideas and guide longer-term change. Post-occupancy evaluation[41] remains one of the few established instruments that redirects attention to places after implementation, yet it is primarily an expert-led, evaluative exercise, typically conducted once rather than continuously, and its findings rarely inform ongoing participatory processes.

Although participatory practices have diversified considerably over time, many continue to share similar underlying temporal assumptions. Table 2 provides an overview of the historical evolution of major participatory methods and practices. Despite their diversity, most participatory methods share a common characteristic: they are generally organized around specific projects, interventions, or decision-making moments. Participation typically begins at a defined point and concludes once plans have been approved or projects have been implemented. Consequently, the majority of participatory practices remain temporally bounded and rarely address the long-term evolution of places or the changing needs of users.

This observation does not diminish the importance of existing participatory methods. Public meetings, charrettes, workshops, co-design exercises, and deliberative forums continue to play important roles within democratic planning and design processes. However, understanding

Table 2. Historical evolution of participatory methods

Method	Approximate origins	Main purpose
Public hearings	Early 20th century	Information and consultation
Public meetings	Mid-20th century	Citizen feedback
Advocacy planning	1960s	Representation and social justice
Community workshops	1970s–1980s	Collective problem solving
Charrettes	1980s–1990s	Collaborative urban design
Participatory design	Scandinavian tradition, 1970s	User-centered co-design
Co-production and living labs	2000s	Innovation and experimentation
Citizens' assemblies	2000s–2010s	Deliberative democracy
Digital participation	2010s	Expanded accessibility and engagement
Participatory temporal urbanism mechanisms	Emerging	Longitudinal participation and place evolution

participation exclusively through such methods risks overlooking the ongoing relationships that emerge after implementation and continue throughout the life cycle of places.

The challenge, therefore, is not to replace existing methods but to reconsider the temporal assumptions embedded within them. Rather than viewing participation as a discrete event, there is a need to explore how participation can extend beyond individual projects and become embedded within processes of stewardship, adaptation, and long-term place evolution. This shift provides the basis for the concept of PTD introduced in the following section.

5. Designing for time: Toward participatory temporal design

Participation and design have historically been closely interconnected. From community architecture and participatory design traditions to contemporary co-design practices, scholars and practitioners have increasingly emphasized the importance of involving users in shaping the environments they inhabit.[4,21,22] These approaches have contributed significantly to democratizing design processes and challenging traditional distinctions between experts and users. At the same time, many participatory design practices remain closely associated with specific projects and relatively short time horizons. Co-design workshops, charrettes, consultations, and participatory exercises are typically organized around clearly defined interventions and conclude once implementation is complete. Although such approaches improve inclusiveness and design quality, they often assume relatively stable user groups and fixed needs. However, urban environments are rarely static. Places evolve through everyday practices, demographic change, shifting lifestyles, technological transformations, and changing cultural meanings. Users themselves change over time as they age, move, form new relationships, or experience changing circumstances. Consequently, the social life course of places frequently extends far beyond the temporal boundaries of individual projects.

Recent scholarship on maintenance, repair, care, and stewardship suggests that the long-term life cycle of places may be as important as their initial design.[7,8] Rather than understanding design as a completed product, these perspectives emphasize ongoing processes of maintenance, adaptation, and collective responsibility. Similarly, commons scholarship demonstrates that communities can manage shared resources through long-term cooperation and evolving forms of governance.[28]

Building on these perspectives, this study proposes the notion of PTD. PTD refers to design approaches that recognize change, adaptation, and long-term stewardship as integral dimensions of participation. Rather than focusing exclusively on the creation of places, PTD emphasizes their continuous evolution and the changing relationships between users and their environments over time. PTD rests on four interrelated principles. First, design should acknowledge that users and communities are dynamic

rather than fixed entities. Second, participation should extend beyond project implementation and support long-term stewardship and adaptation. Third, design processes should accommodate uncertainty and allow places to evolve in response to changing circumstances. Finally, intergenerational continuity and collective memory should be recognized as important dimensions of sustainable place development.

Table 3 summarizes the relationship between conventional participatory design, PTD, and PTU.

Participatory temporal design should therefore be understood as the design-oriented expression of PTU rather than as a separate theoretical framework. Whereas PTD focuses on how designers, planners, and communities can create places that accommodate change over time, PTU provides the broader conceptual framework through which these evolving relationships among people, places, institutions, and urban change can be understood. In this sense, PTD operates at the scale of design practice, while PTU offers a wider theoretical perspective on the long-term temporal life of places.

These principles do not replace existing participatory methods but rather complement them by introducing a temporal perspective. Public meetings, workshops, charrettes, and co-design processes remain important instruments within democratic planning and design. However, they should be understood not as isolated events but as moments within broader, ongoing relationships among people, places, and time. PTD thus provides a bridge between established traditions of participatory planning and the broader framework of PTU introduced in the following section.

6. Participatory temporal urbanism

Building upon these traditions, this study proposes PTU as a conceptual framework for understanding participation as an evolving relationship rather than a singular event. PTU recognizes that cities, places, and communities are continuously changing and that participation should therefore extend beyond individual projects and remain embedded within longer processes of stewardship, adaptation, and collective memory.

Rather than replacing existing participatory approaches, PTU seeks to complement and extend them by introducing time as a central analytical category. In this sense, PTU aligns with broader calls for theoretical pluralism and relational understandings within urban design and planning scholarship.[42] The framework emphasizes five interconnected dimensions: co-creation, deliberation, co-management, co-adaptation, and co-memory.

Participatory temporal urbanism is therefore offered neither as a wholly new invention nor as a relabeling of familiar ideas, but as a deliberate synthesis with an original theoretical contribution: it takes time as the central analytical category of participation theory and specifies, through five interrelated dimensions, what participation consists of once its temporal limitations are reconsidered.

The PTU framework extends participation theory by connecting it to the post-implementation literatures identified above. It fills the gap those literatures leave by keeping citizens, rather than managers or institutions, at the center of the account, and it moves beyond the truism that "participation should continue after implementation" by specifying, for each dimension, the actors, mechanisms, temporal scales, institutional requirements, and risks involved. The contribution is, in this sense, synthetic: the elements are known, but the building is new. The deeper proposal of this study is that time can serve as an organizing principle for participation theory, a category through which its existing insights on power, communication,

Table 3. Relationship between conventional participatory design, participatory temporal design, and participatory temporal urbanism

Dimension	Conventional participatory design	Participatory temporal design	Participatory temporal urbanism
Primary focus	Design of projects and places	Design for long-term change and adaptation	Long-term evolution of participation across the life of places
Time horizon	Mainly project-based	Project plus post-implementation	Continuous across generations
Main concern	User involvement in design	Designing for changing users and places	Participation as an ongoing relationship
Scale	Project and site	Project and neighborhood	Urban systems, governance and place evolution
Main outcome	Better design solutions	Adaptable and resilient places	Long-term stewardship, adaptation and co-memory

co-production, and care can be re-read and reconnected. PTU is offered as a first framework built on that principle, not as its final form.

Conventional approaches to participation have generally been organized around project-based and linear processes in which engagement is concentrated at the beginning of planning and gradually diminishes after implementation. As illustrated in Figure 1, participation is often treated as a finite sequence with clearly defined starting and ending points.

By contrast, PTU proposes an alternative understanding based on continuity, adaptation, and evolving relationships between communities and places. As illustrated in Figure 2, the five dimensions of PTU should be understood as interconnected, cyclical processes rather than sequential stages. Participation is therefore conceived not as a temporary intervention but as an ongoing relationship embedded within the temporal life of places.

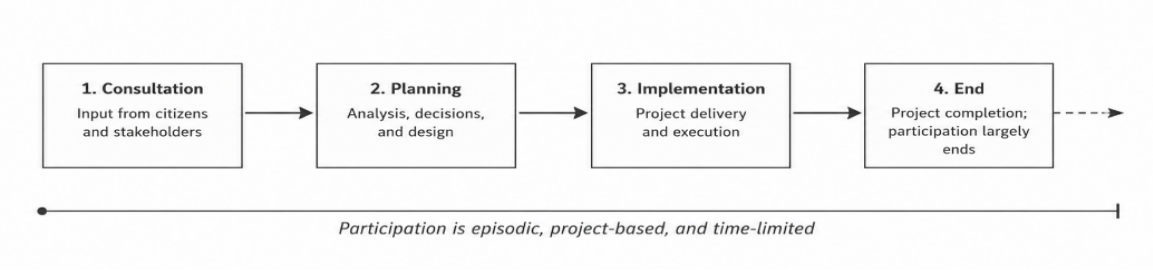


Figure 1. Conventional participation model. The conventional participation model is embedded in a linear, project-based logic. Participation typically occurs during consultation and planning but gradually diminishes as implementation progresses. Once projects are completed, formal participation generally ends, despite the continuing evolution of places and communities.

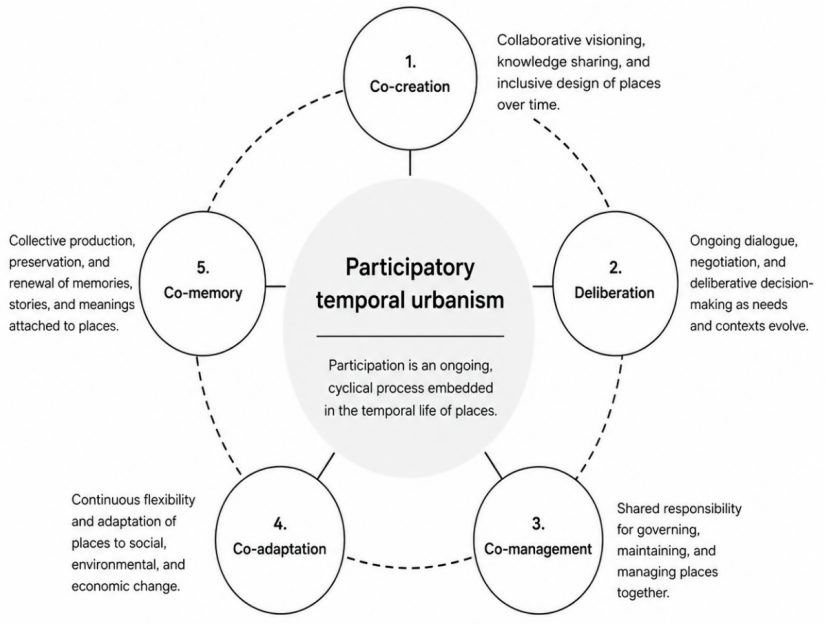


Figure 2. Participatory temporal urbanism framework. Participatory temporal urbanism reconceptualizes participation as an ongoing, cyclical process embedded in the temporal life of places. Through co-creation, deliberation, co-management, co-adaptation, and co-memory, communities continually shape, care for, and renew their environments. Participation therefore evolves over time rather than ending with implementation. The five dimensions operate conjointly rather than sequentially; the circular arrangement indicates interrelation, not chronological order.

6.1. Co-creation

Co-creation is the most familiar and widely recognized dimension of participation and closely aligns with many existing traditions in participatory planning and participatory design. Workshops, public meetings, charrettes, co-design exercises, visioning processes, and collaborative forums all provide opportunities for users and communities to contribute to the shaping of places before implementation. Through such mechanisms, local knowledge, lived experience, and diverse perspectives can be incorporated into planning and design processes.

Within conventional participatory approaches, co-creation is often regarded as the primary objective of participation. The emphasis is typically placed on generating ideas, identifying needs, improving design quality, and strengthening democratic legitimacy. Participatory design traditions have similarly emphasized users as active contributors rather than passive recipients of expert knowledge.[21-23] Co-creation therefore represents one of the most important achievements of contemporary participatory thinking.

At the same time, PTU argues that co-creation should not be understood as the endpoint of participation but rather as its initial stage. Although the initial design and planning phases are crucial, the social life of places continues long beyond implementation. Users change, communities evolve, and new needs emerge over time. Consequently, participation cannot be confined solely to moments of creation.

Within PTU, co-creation remains essential but is understood as the beginning of a much longer relationship between people and places. Initial participation lays the foundation for subsequent processes of deliberation, stewardship, adaptation, and collective memory to unfold. In this sense, co-creation represents not a final outcome but the first stage in the ongoing temporal evolution of places.

As summarized in Table 4, co-creation is the foundation of PTU, but its long-term effectiveness depends on inclusive participation mechanisms that support ongoing processes of deliberation, stewardship, adaptation, and collective memory.

6.2. Deliberation

Participation involves not only expression but also reflection and collective judgment. Deliberative approaches

Table 4. Operational dimensions of participatory temporal urbanism

PTU dimension	Primary purpose	Main actors	Typical mechanisms	Temporal scale	Institutional considerations	Potential risks
Co-creation	Shape places collaboratively	Citizens, planners, designers, and stakeholders	Workshops, charrettes, co-design, participatory planning	Planning and design phase	Inclusive participation processes	Tokenism, exclusion of marginalized voices
Deliberation	Collective learning and decision-making	Citizens, public authorities, and experts	Citizens' assemblies, jury processes, collaborative forums	Throughout planning and adaptation	Transparent governance and facilitation	Dominance of vocal interests, unequal influence
Co-management	Long-term stewardship and maintenance	Municipalities, community organizations, and users	Stewardship agreements, maintenance partnerships, commons governance	Long-term	Stable governance arrangements and funding	Volunteer fatigue, institutional discontinuity
Co-adaptation	Respond to changing needs and conditions	Communities, planners, and local authorities	Monitoring, feedback, adaptive management, incremental interventions	Continuous	Flexible regulations and learning systems	Resistance to change, fragmented responsibility
Co-memory	Preserve and reinterpret collective meanings	Residents, heritage groups, and cultural institutions	Heritage interpretation, storytelling, commemorations, everyday practices	Generational	Heritage recognition, archives, cultural programming	*Exclusionary or nostalgic narratives; contested memories*

Abbreviations: PTU: Participatory temporal urbanism.

emphasize dialogue, learning, and the consideration of multiple perspectives. Drawing upon traditions of communicative and collaborative planning, deliberation seeks to create spaces where different interests, values, and forms of knowledge can engage in constructive interaction. Contemporary examples include citizens' assemblies, collaborative planning forums, participatory design processes, and other forms of collective decision-making intended to strengthen democratic legitimacy and mutual learning.

Particular attention may also be given to the jury principle associated with New Urbanism and the work of Duany[43,44] Inspired by legal jury systems, this approach seeks to balance expertise, evidence, and citizen deliberation while avoiding the domination of participatory processes by the loudest voices or the most organized interests.[27,43,44] Duany[27] has argued that conventional public meetings frequently privilege highly motivated local interests rather than the broader community and has proposed more representative forms of participation based on randomly selected citizen groups resembling jury systems. Rather than assuming that participation automatically produces consensus, the jury principle emphasizes structured deliberation, informed discussion, and the careful weighing of competing perspectives. Within New Urbanist practice, charrettes have frequently served as arenas for such collective learning and negotiation.[25-27]

From a PTU perspective, deliberation also possesses a temporal dimension. Decisions made at one moment inevitably influence future users and communities who are not present during the original planning process. Participation, therefore, involves not only current stakeholders but also responsibilities toward future generations and the long-term evolution of places. Deliberation becomes not merely a mechanism for achieving agreement in the present but a means of considering future consequences and intergenerational continuity.

Deliberation thus represents an important mechanism through which participation can achieve legitimacy, collective learning, and temporal continuity. Within PTU, deliberation is understood not as a single event but as an ongoing process through which communities continuously negotiate, reinterpret, and adapt the meanings and futures of places. Its effectiveness, however, depends on transparent institutional arrangements that ensure equitable participation and prevent the concentration of influence among already empowered actors.

6.3. Co-management

Participation does not necessarily end once a project has been completed. Communities frequently continue to care for, manage, and maintain places through formal and informal stewardship practices. Commons scholarship and research on stewardship demonstrate that users often assume long-term responsibilities for shared environments. Co-management recognizes maintenance, repair, and collective responsibility as integral dimensions of participation. In this perspective, participation becomes closely associated with practices of care rather than merely consultation. The ongoing life of places depends not only on their initial design but also on their continued management and everyday use. Long-term co-management therefore requires stable institutional support, adequate resources, and shared responsibility between municipalities and communities for stewardship to remain viable over time.

6.4. Co-adaptation

Places evolve over time in response to changing demographic conditions, social practices, technological transformations, and cultural meanings. Similarly, users themselves change over the course of their lives. Consequently, participation must accommodate uncertainty and support continuous adaptation. Co-adaptation acknowledges that no design can anticipate every future need. Instead, sustainable places should possess the capacity to evolve through incremental change and continuous feedback processes. Participation therefore becomes a mechanism for enabling responsiveness rather than preserving fixed solutions. Adaptive participation consequently depends upon governance arrangements that encourage learning, flexibility, and the periodic reconsideration of earlier planning decisions.

6.5. Co-memory

Places are repositories of collective memories, identities, and shared experiences. Through everyday practices and collective narratives, communities continually reinterpret and assign meaning to urban environments. Participation therefore extends beyond physical interventions and includes the preservation, transformation, and transmission of social and cultural memories.

Unlike conventional understandings of participation, which focus primarily on decision-making and project implementation, co-memory emphasizes memory itself as a participatory practice. Communities actively construct, reinterpret, and reproduce meanings associated with places through stories, rituals, everyday practices, commemorations, and informal interactions over time. In this sense, memory is not simply inherited from the past but continually recreated in the present.

Co-memory also highlights the intergenerational dimensions of participation. Places are shaped not only by present users but also by accumulated experiences,

inherited traditions, and future expectations. Memories become embedded in streets, buildings, landscapes, and public spaces, creating continuity that connects different generations. Participation therefore includes both the preservation and reinterpretation of memories. Among the five dimensions, co-memory represents one of the least explored dimensions of participation theory. As illustrated in Figure 3, memory itself can be understood as a participatory practice through which stories, rituals, everyday practices, heritage, and intergenerational transmission continually shape and reshape the meanings associated with places.

Questions of heritage and attachment further illustrate the importance of co-memory. Communities frequently attribute symbolic meanings to places that extend beyond their physical characteristics. Such meanings evolve over time and are constantly renegotiated through everyday use and collective experience. Co-memory therefore recognizes that places possess temporal depth and that sustainable urban development requires sensitivity not only to future needs but also to existing and inherited meanings and identities. By introducing co-memory into participation theory, PTU expands conventional understandings of engagement beyond consultation and decision-making. Participation becomes a process through which communities continuously reproduce, preserve, reinterpret, and transform the meanings associated with places. Together, the five dimensions of PTU—co-creation, deliberation, co-management, co-adaptation, and co-memory—illustrate how participation may be understood as an ongoing process embedded within the temporal life of places. Participation thus becomes not merely a means of achieving project legitimacy but a continuing relationship linking people, environments, and urban change across time. PTU does not reject existing participatory methods; rather, it proposes that participation should be understood through broader temporal horizons that acknowledge stewardship, adaptation, and memory as essential dimensions of sustainable urban development. Co-memory also requires conscious attention to whose histories, identities, and experiences are preserved, ensuring that collective memory remains inclusive rather than privileging dominant narratives alone.

Taken together, these five dimensions illustrate that participation is best understood not as a sequence of isolated planning activities but as an evolving temporal relationship that accompanies the life of places. Rather than functioning as rigid stages, they represent interconnected and mutually reinforcing processes through which places are co-created, governed, adapted, and continually reinterpreted over time. This integrative conceptual framework provides the foundation for interpreting the illustrative examples presented in the following section.

7. Illustrative examples of participatory temporal urbanism

Participatory temporal urbanism is intended primarily as a conceptual framework rather than an empirical model. Nevertheless, many cities and places already exhibit characteristics that resonate with its underlying principles. Although these examples were not explicitly designed in accordance with PTU, they demonstrate how participation, adaptation, stewardship, and collective memory often unfold over extended periods.

Piazza Navona in Rome (Classical European Square; historical continuity) illustrates how places evolve through successive layers of history, meaning, and everyday use. Originally constructed on the ruins of the Stadium of Domitian, the square has continually adapted to changing political, cultural, and social contexts throughout nearly two millennia (Figure 4). Its present identity reflects not a single moment of creation but an ongoing process of reinterpretation and collective memory. Through festivals, everyday encounters, tourism, and civic life, Piazza Navona continues to acquire new meanings while preserving traces of its historical past. From a PTU perspective, the square demonstrates how continuity and change can coexist across generations. Although Piazza Navona was not created through contemporary participatory planning processes, it illustrates how long-term place evolution and collective memory can contribute to the formation of participatory meanings through continuous everyday appropriation across generations.

Similarly, the case of Gamla Stan in Stockholm (The Old City of Stockholm; incremental adaptation) demonstrates the importance of gradual change and continuity. Rather than remaining fixed, the historic district has continuously accommodated new functions, residents, visitors, and economic activities while preserving its distinctive identity (Figure 5). The area illustrates how adaptation and stewardship can coexist with heritage conservation and place attachment. Through everyday use, preservation efforts, and changing social practices, Gamla Stan has evolved without losing its historical character. The district exemplifies how places may retain their identity while responding to evolving urban realities over time.

Tempelhofer Feld in Berlin (Transformed Former Airport; open-ended stewardship) represents another example of evolving participation and adaptive place-making processes. Following the closure of the former Tempelhof Airport, citizens successfully resisted

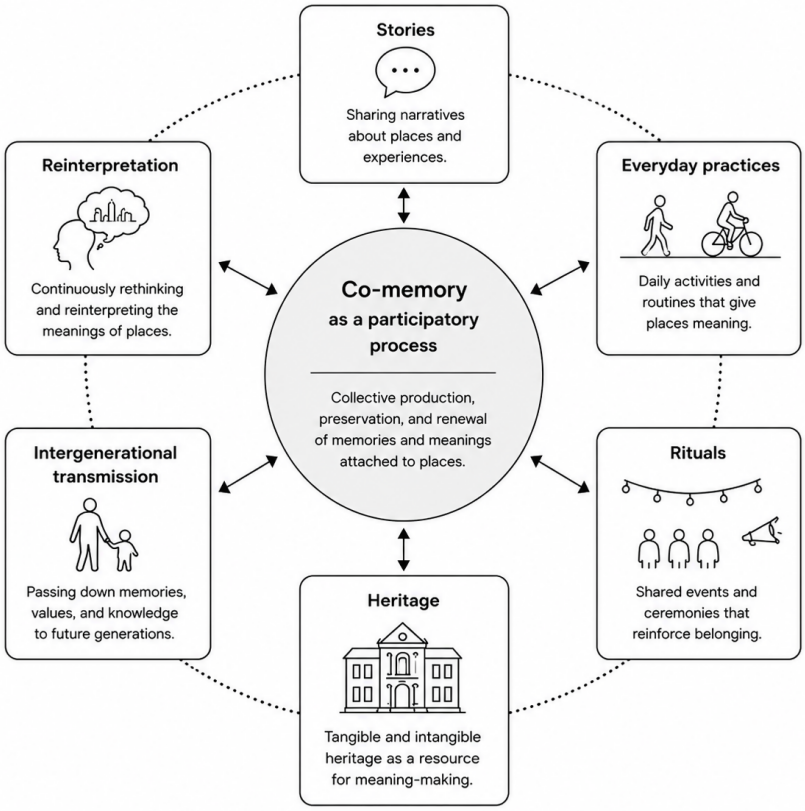


Figure 3. Co-memory as a participatory process. Co-memory recognizes memory itself as a participatory practice. Through stories, rituals, everyday use, heritage, reinterpretation, and intergenerational transmission, communities continually reproduce and transform the meanings associated with places. Participation therefore extends beyond physical interventions and includes the collective production, preservation, and renewal of memory.



Figure 4. Illustrative example of participatory temporal urbanism: Piazza Navona, Rome, Italy. Source: DjedJ, Marseille, France (Pixabay). Used under the Pixabay Content License.



Figure 5. Illustrative example of participatory temporal urbanism: Gamla stan, Old Town, Stockholm, Sweden. Source: Sharon Ang, Singapore (Pixabay). Used under the Pixabay Content License.

conventional redevelopment proposals and contributed to preserving the site as an open public landscape (Figure 6). Since then, the space has continued to evolve through everyday use, temporary interventions, community gardens, cultural events, and changing forms of social appropriation. Rather than following a fixed master plan, Tempelhofer Feld illustrates how openness, stewardship, and collective negotiation can shape the long-term life of places. The project demonstrates how participation can continue after formal planning processes conclude, allowing users to continually redefine the meaning and functions of urban space.

The High Line in New York (Elevated Landscape Park; continued adaptation after implementation) demonstrates how participation and stewardship can extend beyond the initial design phase. Originally initiated through civic activism and community advocacy, the project transformed an abandoned elevated railway into one of the internationally recognized public spaces (Figure 7). Since its completion, the High Line has continued to evolve through programming, maintenance, cultural activities, and changing patterns of use. While often discussed in relation to regeneration and gentrification, the project also illustrates how public spaces acquire new meanings over time through continued interaction among institutions, users, and local communities. From a PTU perspective, the High Line exemplifies participation as an ongoing process rather than a finite intervention. At the same time, the High Line also demonstrates that long-term participation may generate unintended consequences, including gentrification, tourism pressures, and changing patterns of social inclusion. From a PTU perspective, such tensions highlight the importance of ensuring that long-term stewardship is accompanied by equity-oriented governance arrangements.

More contemporary examples can be found in projects such as Copenhagen's Superkilen (Multicultural Urban Public Space; extreme participation and evolving meanings). Developed through extensive community engagement, the project sought to reflect the cultural diversity of its surrounding community. Objects and symbols representing numerous countries were incorporated into the design, creating a public space intended to represent multiple identities and experiences (Figure 8). Since its completion, however, the meanings associated with the space have continued to evolve through everyday practices, changing demographics, and new forms of user appropriation. Superkilen demonstrates that participation does not end with design but continues as communities reinterpret and inhabit urban spaces over time. The project also raises broader questions concerning representation, cultural symbolism, and whether participatory design alone is sufficient to sustain meaningful long-term engagement after implementation. These issues reinforce the importance of viewing participation as an evolving process rather than a completed design achievement.

Jubileumsparken in Gothenburg (The Jubilee City Park; prototyping and long-term experimentation)



Figure 6. Illustrative example of participatory temporal urbanism: Tempelhofer Field, Berlin, Germany. Source: P.S. Greiner, Berlin, (Pixabay). Used under the Pixabay Content License.



Figure 7. Illustrative example of participatory temporal urbanism: The High Line Park, New York City, United States. Source: Shire777, USA New York (Pixabay). Used under the Pixabay Content License.

provides another example of long-term and experimental participation. Temporary interventions, iterative design processes, and ongoing community involvement have allowed the project to evolve gradually rather than according to a fixed master plan. Swimming facilities, public spaces, and cultural activities were introduced incrementally, allowing users to influence the park's ongoing development over time (Figure 9). The project illustrates how participation can extend beyond conventional planning stages and remain embedded within processes of adaptation, experimentation, and collective stewardship. From a PTU perspective, Jubileumsparken demonstrates how uncertainty itself can become a productive condition of urban development.

These six illustrative examples suggest that many successful places are characterized not by permanence

or finality but by their capacity to evolve. From a PTU perspective, the significance of participation lies not only in the creation of places but also in their continued reinterpretation, management, and renewal over time.

Table 5 demonstrates that the illustrative examples do not represent identical manifestations of PTU. Rather, each highlights different dimensions of the framework and illustrates how participation may unfold in distinct ways across different historical, institutional, and cultural contexts. Together, the examples suggest that PTU should not be understood as a fixed model to be replicated but rather as a conceptual framework for interpreting diverse trajectories of place evolution, stewardship, adaptation, and memory formation.

8. Discussion

The principal contribution of PTU is not to replace existing theories of participation but to integrate them within a common temporal framework. Whereas previous scholarship has generated important insights into participation, collaborative planning, participatory design, stewardship, commons governance, care, maintenance, and adaptive management, these perspectives have generally evolved as parallel conversations. PTU proposes that time provides a previously underexplored integrative dimension that reconnects these traditions into a more coherent understanding of how participation unfolds across the life of places. Rather than conceptualizing participation as a sequence of isolated planning events, PTU understands participation as a continuous relationship linking people, places, institutions, and urban change over time.

From a planning perspective, the framework highlights the importance of moving beyond project-based understandings of participation. Public hearings, workshops, charrettes, and collaborative processes remain valuable instruments for engaging communities and generating local knowledge. However, sustainable urban development may also require mechanisms that support long-term stewardship, adaptation, and shared responsibility. Participation therefore becomes not merely a tool for legitimizing projects but a continuing relationship between communities and places.

The framework also has implications for design practice. Participatory design has traditionally focused on involving users in shaping environments and products. PTD extends this perspective by recognizing that users, communities, and place meanings change over time. Consequently, design should not be understood solely as the production of finished objects or spaces but as the creation of conditions that allow for future adaptation and reinterpretation.

Extending participation across time, however, raises practical and institutional questions that the framework itself cannot fully resolve. Co-management and stewardship of the kind illustrated by Tempelhofer Feld or Jubileumsparken depend on sustained funding, organizational continuity, and legal or administrative structures that outlast individual projects, budget cycles, and changes in municipal leadership. Foster and Iaione[45] have argued that durable urban commons typically require dedicated governance arrangements, such as collaborative pacts or co-management agreements between residents and municipalities, rather than relying on goodwill or informal continuity alone. Without such structures, the



Figure 8. Illustrative example of participatory temporal urbanism: Superkilen, Kileparken, Copenhagen, Denmark. Source: Bente Jønsson, Søborg, Danmark (Pixabay). Used under the Pixabay Content License.



Figure 9. Illustrative example of participatory temporal urbanism: Jubileumsbadet (the Jubilee Baths/Pool) in Jubileumsparken, Gothenburg, Sweden. Source: Author-provided image.

Table 5. Illustrative assessment of the six participatory temporal urbanism examples

Case	Co-creation	Deliberation	Co-management	Co-adaptation	Co-memory	Governance/key lesson
Piazza Navona	Low (historic origin)	Moderate	High	High	Very high	Demonstrates long-term place evolution and accumulated collective memory
Gamla Stan	Low	Moderate	High	High	Very high	Shows incremental adaptation while preserving historic identity
Tempelhofer Feld	High	High	High	Very high	Moderate	Illustrates adaptive stewardship through continued civic involvement
The High Line	High	Moderate	High	High	Moderate	Demonstrates long-term adaptation but also reveals tensions related to regeneration and equity
Superkilen	High	High	Moderate	High	High	Highlights multicultural participation while raising questions about representation and symbolic inclusion
Jubileumsparken	Very high	High	High	Very high	Moderate	Shows how institutionalized experimentation and prototyping can sustain participation through uncertainty

long-term participation envisioned by PTU risks remaining aspirational rather than institutionally embedded, particularly in municipalities with limited administrative capacity or constrained budgetary priorities.

A related concern is equity. The critical literature on participation and power[14,16,17] discussed in Section 3.1 suggests that sustained, long-term involvement is not equally accessible to all residents. Co-creation workshops or public consultations can, at least in principle, be designed to reach a broad cross-section of a community within a defined timeframe. Co-management and co-memory, by contrast, depend on the kinds of ongoing time commitments, social capital, and institutional access that are unevenly distributed. There is therefore a risk that PTU, if implemented without explicit attention to inclusion, could reinforce rather than redistribute existing power asymmetries, allowing already engaged residents to accumulate stewardship roles and decision-making influence over time while less-resourced groups participate only during the initial co-creation stage. Addressing this risk would require deliberate mechanisms, such as rotating stewardship roles, supported participation, or institutional support for under-resourced groups, to ensure that long-term participation does not become the preserve of those with the most time and resources to sustain it.

The growing literature on care, maintenance, and commons governance provides additional theoretical support for such a perspective. Scholars have increasingly emphasized that the long-term success of places depends not only on innovation and construction but also on practices of repair, stewardship, and collective responsibility. Gabauer *et al.*[33] bring together cross-disciplinary accounts of how care and uncare are unevenly distributed across urban space, while Davis[34] develops an explicit ethics of care for urban design, arguing that the built environment itself can be shaped to support relations of interdependence, capability, and long-term wellbeing rather than treating care as something introduced after design is complete. In this sense, participation becomes closely associated with care and maintenance rather than being confined to consultation and decision-making. PTU also contributes to contemporary debates on sustainability by expanding sustainability beyond its environmental, economic, and social dimensions to include a distinctly temporal dimension. Sustainable places are not simply those that perform efficiently at a single point in time, but those that adapt to changing environmental conditions, demographic transitions, technological change, cultural transformation, and evolving community needs without losing their social or democratic vitality. In this sense, PTU contributes simultaneously to social sustainability through continued participation and inclusion, institutional sustainability through long-term stewardship and governance, environmental sustainability through adaptive place management, and intergenerational sustainability through the preservation and reinterpretation of collective memory.

The concept of place evolution introduced in this study challenges static understandings of place-making and place preservation. Places are neither fixed products nor entities that can be maintained unchanged. Rather, they continuously evolve through everyday practices, changing populations, and shifting cultural meanings. Understanding this process may help planners and designers better respond to uncertainty and complexity.

From a practical perspective, PTU also suggests the need for institutional adaptation and innovation. If participation is to continue throughout the life of places, municipalities and other planning authorities require governance arrangements that support long-term stewardship rather than limiting participation to project approval stages. Such mechanisms may include neighborhood stewardship partnerships, municipal co-management agreements, long-term participatory budgeting, community land trusts, public-space trusts, periodic post-occupancy participation programs, and rotating forms of community representation designed to ensure that participation remains open to new residents and to changing communities over time. Rather than prescribing a single institutional model, PTU highlights the importance of creating governance structures that support participation as an ongoing process of urban governance.

Several limitations should also be acknowledged. PTU is presented here as a conceptual framework rather than an empirically tested model, and the illustrative examples discussed in Section 7 are drawn primarily from prominent, well-resourced public spaces in European and North American urban contexts. It remains an open question whether the framework's five dimensions operate in the same way in smaller, less visible places, in cities with less established planning institutions, or in informal and resource-constrained settlements where municipal capacity for long-term co-management may be limited or unevenly distributed.

Future research could explore how the dimensions of PTU operate across different cultural contexts, governance systems, and spatial scales, including comparative case studies of municipalities that have institutionalized long-term stewardship arrangements and empirical investigations of who participates and who is excluded as participation extends across time. The framework should also be tested empirically across a broader range

of urban contexts, including cities in the Global South, informal settlements, shrinking cities, and resource-constrained municipalities, where governance capacities and institutional conditions differ substantially from those illustrated in this study. As scholars of Southern urbanism have argued, participation theory does not always translate effectively into contexts of informality and constrained state capacity,[46] where long-term stewardship is often sustained through insurgent and informal practices rather than formal institutions.[47]

Ultimately, PTU does not seek to replace existing theories of participation. Instead, it proposes that time itself deserves greater attention within planning and design theory. By emphasizing continuity, adaptation, and collective memory, while remaining attentive to the institutional and equity challenges of sustaining participation over time, the framework offers a possible avenue for rethinking participation in an era characterized by uncertainty, complexity, and ongoing urban change.

9. Conclusion

Participation has become one of the most influential concepts in planning and design theory. Over the past several decades, scholarship has expanded from questions of power and representation to collaborative planning, participatory design, co-production, and more recent discussions concerning care, stewardship, and commons governance. Despite these advances, participation continues to be understood predominantly through project-based and episodic frameworks.

This study has argued that time remains one of the least explored dimensions of participation theory. Building upon insights from planning theory, participatory design, care studies, commons scholarship, and temporal perspectives in urban studies, the study proposes PTU as a conceptual framework for understanding participation as an evolving relationship rather than a singular event.

The framework identifies five interconnected dimensions—co-creation, deliberation, co-management, co-adaptation, and co-memory—which collectively emphasize continuity, stewardship, adaptation, and collective memory as integral dimensions of sustainable urban development. Rather than replacing existing approaches, PTU seeks to extend them by introducing broader temporal horizons into planning and design theory and practice. As discussed in Section 8, however, realizing this extension in practice depends on institutional arrangements capable of supporting participation over time and on deliberate attention to who is able to participate as that time horizon lengthens.

The study also proposed PTD as an intermediary perspective that links established participatory methods to the longer-term evolution of places. In doing so, it highlighted the importance of viewing cities not as finished products but as continuously evolving environments shaped by changing users, practices, and meanings.

For planners, designers, and policymakers, these arguments carry several practical implications. Participation budgets and timelines designed around a single project milestone may need to be reconsidered in light of longer-term funding and governance mechanisms that support stewardship beyond implementation. Institutional roles, such as designated stewardship organizations, community land trusts, or co-management agreements, may be needed to carry participatory responsibility across the political and budgetary cycles that conventional planning processes rarely survive. At the same time, such mechanisms should be designed with explicit attention to inclusion, so that extended participation has the potential to broaden rather than narrow who shapes the long-term life of places. Ultimately, sustainable cities are not merely designed, implemented, and regarded as complete. They are continually interpreted, maintained, adapted, and renewed through ongoing relationships between people and places. Recognizing these temporal dimensions may help planners, designers, and communities move beyond exclusively project-based understandings of participation and toward more resilient, caring, and evolving forms of urban development.

Future research may further explore the empirical applications of PTU, including longitudinal case studies of municipalities that have institutionalized long-term co-management arrangements, comparative analyses of governance models for sustaining stewardship across political cycles, and empirical investigations of how access to long-term participation is distributed across different communities. By bringing time, along with the institutional and equity questions that accompany it, to the forefront of debates on participation, the framework proposed in this study represents a possible contribution to the ongoing evolution of planning and design theory. PTU argues that participation is not merely a phase of planning but a condition of urban life. By placing time at the center of participation theory and integrating previously fragmented traditions into a coherent conceptual framework, PTU invites planners, designers, researchers, and policymakers to rethink participation as a continuous process of co-creation, stewardship, adaptation, and collective memory through which places evolve over time.

Acknowledgments

I would like to thank the colleagues and scholars who have contributed to ongoing discussions on urban theory,

participation, and place evolution. I am particularly grateful for the exchanges and insights shared during the Future of Places Conference (FOP I–III), the Congress for New Urbanism (CNU), and the Athena Distinguished Female Scholars Series. I am especially indebted to Charles Bohl, Andres Duany, Elizabeth Plater Zyberk, Michael Mehaffy, Emily Talen, David Brain, Goran Cars, Christine Boyer, Fran Tonkiss, Karen Franck, Setha Low, Matthew Carmona, Ali Madanipour, Vikas Mehta, Charles Wolfe, Dana Cuff, Daniel Solomon, Peter Calthorpe, Rachelle Alterman, and the late Judith Innes. Their insights, discussions, and intellectual exchanges have helped shape many of the ideas developed in this conceptual study.

Funding

The author received no specific funding for this research.

Conflict of interest

The author declares no conflict of interest.

Author contributions

This is a single-authored article.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data

No datasets were generated or analyzed during the current study.

Further disclosure

Artificial intelligence tools, including OpenAI ChatGPT, Anthropic Claude, and Perplexity AI, were used during manuscript preparation to assist with language refinement, editing, reference checking, and the exploration and organization of ideas. These interactions functioned similarly to discussions with colleagues or academic peers and were used to improve clarity and consider alternative perspectives. All original ideas, theoretical contributions, interpretations, arguments, and final editorial decisions were conceived, developed, verified, and approved solely by the author. The author assumes full responsibility for the content of the manuscript and any remaining errors.

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